

RHINELANDER, WI, USA

WMO: 727415

Lat: 45.631N

Lon: 89.482W

Elev: 495

StdP: 95.52

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 04803

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.3	(c) -23.2	(d) -29.9	(e) 0.3	(f) -25.4	(g) -27.3	(h) 0.3	(i) -23.0	(j) 9.2	(k) -8.3	(l) 8.4	(m) -9.0	(n) 2.3	(o) 260	(p) 0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	30.0	21.0	28.3	20.1	27.1	19.4	22.8	27.5	21.7	26.2	20.7	25.0	4.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	17.0	25.0	20.2	15.8	24.1	19.1	14.7	23.3	69.6	27.5	65.8	26.2	62.1	25.3	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.8	(b) 7.8	(c) 7.0	(e) -30.6	(f) 32.7	(g) 1.8	(h) 1.9	(i) -31.9	(j) 34.1	(k) -32.9	(l) 35.2	(m) -33.9	(n) 36.3	(o) -35.2	(p) 37.7		
(a) 8.8	(b) 7.8	(c) 7.0	(e) -30.9	(f) 24.6	(g) 1.7	(h) 1.3	(i) -32.1	(j) 25.6	(k) -33.1	(l) 26.3	(m) -34.1	(n) 27.1	(o) -35.3	(p) 28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-10.2	-8.6	-2.3	5.0	12.2	17.1	19.6	18.2	14.3	7.0	-0.1	-7.3		
	DBStd	11.82	6.91	6.72	6.95	5.54	5.05	4.02	3.38	3.30	4.80	4.97	5.66	6.46		
	HDD10.0	2717	627	522	387	166	34	2	0	0	14	122	304	537		
	HDD18.3	4857	886	755	641	399	200	69	26	42	138	353	552	796		
	CDD10.0	1065	0	0	4	17	102	214	298	256	142	29	2	0		
	CDD18.3	164	0	0	0	1	9	32	66	39	17	1	0	0		
	CDH23.3	1603	0	0	1	8	127	310	661	346	141	9	0	0		
	CDH26.7	382	0	0	0	1	31	70	182	72	26	0	0	0		
(14) Wind	WSAvg	3.1	3.4	3.4	3.3	3.5	3.4	2.9	2.6	2.5	2.8	3.2	3.3	3.3		
(15) Precipitation	PrecAvg	806	28	22	42	66	94	105	101	90	99	77	47	33		
	PrecMax	1078	67	50	89	153	174	178	197	161	222	143	128	75		
	PrecMin	611	5	1	8	17	27	29	23	41	20	22	11	7		
	PrecStd	119	16	12	23	30	35	38	40	34	48	35	24	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	9.8	20.1	24.6	30.1	31.2	32.4	31.1	29.2	24.4	17.1	8.9		
		MCWB	2.5	5.7	13.8	14.6	19.2	22.1	22.9	21.9	21.1	16.8	10.8	6.9		
	2%	DB	2.7	5.8	14.7	21.1	26.9	28.5	30.6	28.6	27.2	21.2	13.1	4.9		
		MCWB	0.9	3.1	9.5	12.1	17.0	19.6	21.7	20.7	19.5	15.7	9.5	3.3		
	5%	DB	1.1	2.8	10.9	17.7	24.1	27.0	28.6	27.0	24.8	18.1	10.8	2.5		
		MCWB	0.1	0.9	6.7	9.5	15.4	19.2	20.7	19.7	18.1	13.5	7.6	1.1		
	10%	DB	-0.8	0.9	7.5	15.0	21.6	24.9	27.2	25.1	22.6	15.5	8.0	1.0		
		MCWB	-1.7	-0.6	4.3	8.3	14.3	17.6	19.7	18.4	17.3	11.7	5.3	0.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	6.5	14.7	16.1	21.2	23.4	25.0	23.7	22.3	19.9	12.7	7.6		
		MCDB	4.5	9.0	18.8	22.1	27.5	28.4	30.1	28.1	26.5	22.1	14.9	8.4		
	2%	WB	1.3	3.1	10.5	13.4	19.0	21.7	23.0	22.3	20.8	16.3	10.3	3.4		
		MCDB	2.5	5.3	13.2	18.9	23.9	26.6	27.9	26.3	25.0	19.8	13.1	4.5		
	5%	WB	0.0	0.7	6.8	10.8	17.1	20.4	21.9	21.1	19.7	14.2	7.6	1.2		
		MCDB	1.1	2.4	10.3	16.3	22.0	24.8	26.7	24.9	23.4	17.6	9.9	2.2		
	10%	WB	-1.6	-1.1	4.1	8.6	15.3	19.1	20.7	20.0	18.3	12.0	5.6	0.1		
		MCDB	-0.7	0.2	7.2	14.7	20.0	23.0	25.3	23.8	21.6	14.8	8.1	0.9		
(35) Mean Daily Temperature Range	MDBR		8.9	10.3	10.8	11.6	12.8	12.1	12.6	12.1	11.9	10.0	8.0	7.9		
	5% DB	MCDBR	9.0	10.5	13.2	16.9	15.8	13.9	14.2	13.5	13.8	13.5	12.4	8.8		
		MCWBR	7.7	8.1	8.4	9.6	7.5	6.9	6.7	6.6	7.1	8.3	8.7	7.3		
	5% WB	MCDBR	8.5	9.8	11.6	14.4	12.8	11.8	12.2	11.4	11.2	11.7	11.0	7.5		
		MCWBR	7.4	7.7	7.8	9.4	6.8	6.7	6.6	6.3	6.6	8.3	8.5	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.240	0.268	0.309	0.335	0.364	0.365	0.362	0.362	0.342	0.306	0.279	0.248		
	taud		2.383	2.297	2.346	2.417	2.377	2.397	2.425	2.425	2.481	2.574	2.541	2.439		
	Ebn at Noon		892	919	923	925	903	901	898	886	878	873	840	844		
	Edh at Noon		71	95	106	109	117	116	111	107	93	73	62	61		
(44) All-Sky Solar Radiation	RadAvg		1.51	2.61	3.82	4.58	5.38	5.82	6.01	5.09	3.96	2.45	1.45	1.07		
	RadStd		0.15	0.28	0.38	0.45	0.45	0.47	0.32	0.33	0.26	0.25	0.15	0.12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page